

Interim progress report

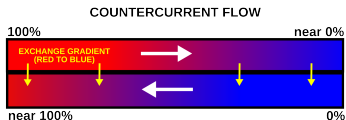
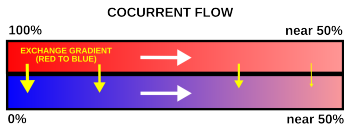
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Project background

Physical models:

- Fluid dynamics (Navier-Stokes eqns.)
- Heat conduction (Heat eqn.)



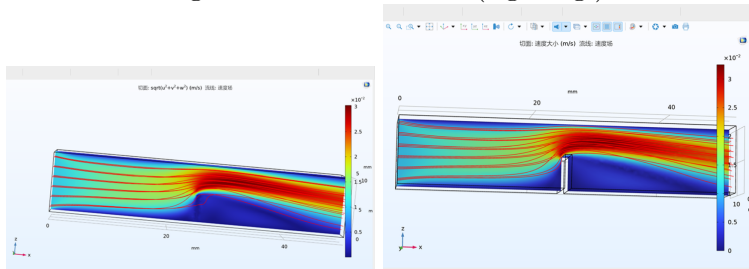
Project background

How is it justified to write an entire project about something so simple and well documented?

- Heat exchangers are traditionally evaluated by use of a body-fitted mesh
- For complex geometries, this is unfeasible
- Warrants the investigation into alternative methods (Brinkman's method)
- We are basing much of the theory on previous work from *Dalian University* and *University of Science and Technology of China*, paper titled *Meshless Optimization of Triply Periodic Minimal Surface Based Two-Fluid Heat Exchanger* by Jiang et al.

Fluid flow

Initially we implemented a simple model for fluid flow using the Brinkman method (left fig.), comparing it to the same geometry evaluated using traditional methods (right fig.).

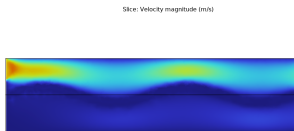
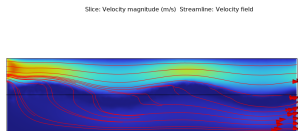


Note: All figures in this presentation are slices from 3D models.

Channel separation

Critical to the performance of the model is that the two channels are adequately separated, i.e. that there is no flow between the two fluids.

Simply implementing the Brinkman method 'as-is' does not separate the channels enough.



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$$\text{Solid region: } |z - 0.05 - 0.01 \sin(10\pi x)| < 0.005$$

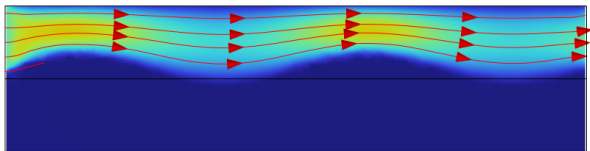
Single channel

For a single channel we can define *both* the wall and other channel so behave like solids, by changing the solid region to:

$$z - 0.05 - 0.01 \sin(10\pi x) < 0.005.$$

This renders:

Slice: Velocity magnitude (m/s) Streamline: Velocity field

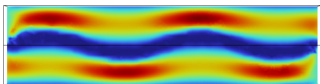


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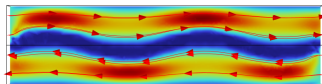
Dual channels

Luckily the Comsol framework can be nicely adjusted to incorporate two channels in the same model:

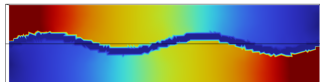
Slice: Velocity magnitude (m/s)



Slice: Velocity magnitude (m/s) Streamline: Velocity field



Slice: Pressure



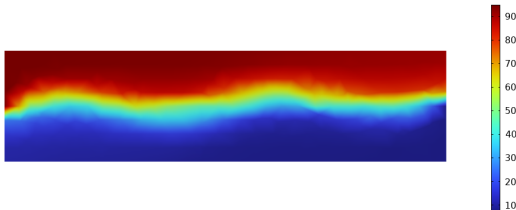
$$z - 0.05 - 0.01 \sin(10\pi x) < 0.005 \quad (\text{Upper channel})$$

$$z - 0.05 - 0.01 \sin(10\pi x) > -0.005 \quad (\text{Lower channel})$$

Heat transfer

Using the same model as before we can turn on the *Heat Transfer in Fluids* module and solve for the temperature distribution. The thermal conductivity is set to $0.6 \text{ W m}^{-1} \text{ K}^{-1}$ for water and $400 \text{ W m}^{-1} \text{ K}^{-1}$ for copper. From this we can see that the current design of the heat exchanger is very poor:

Slice: Temperature (°C)



What is left to do

- Begin investigating more complex geometries
- More thoroughly analysing the results and looking for numerical issues with the models.